

EMP-System Adaptor- Low Level Design

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Version History

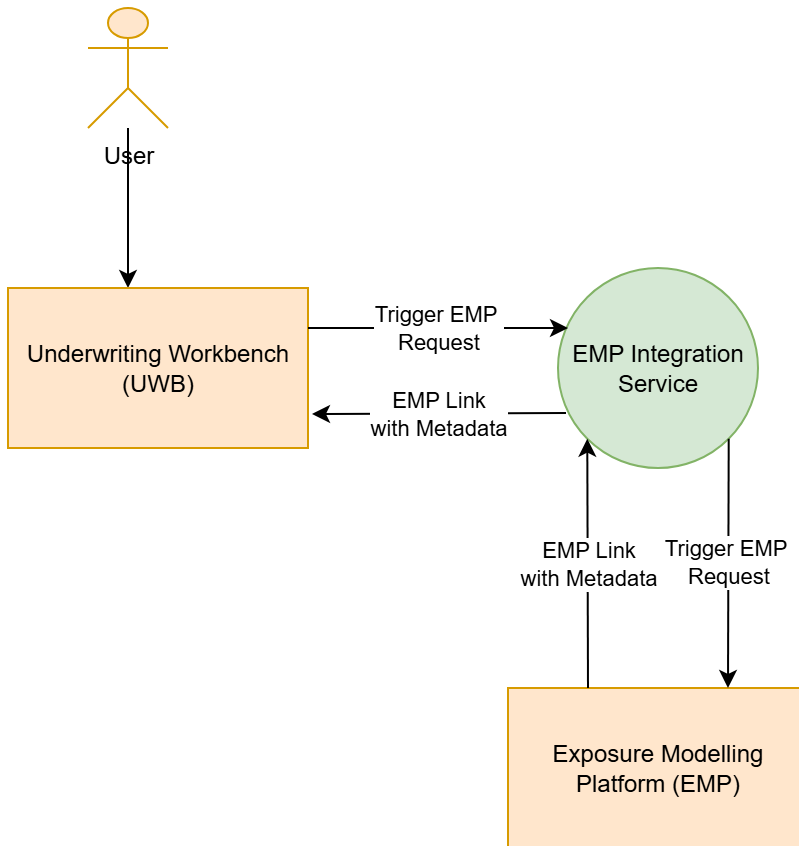
Version	Change	Author	Date
V0.1	Initial creation	Abhishek Bhadgale	05 Jan 2026

Overview

This document provides the Low-Level Design for EMP Integration.

The integration enables Underwriting Workbench to trigger EMP processes directly from UWB without manual data re-entry and to receive process status and EMP links back in UWB. The EMP integration follows Request-Response pattern using Azure Integration components. UWB initiates an EMP request which is routed through Azure API Management and process by Function App before invoking EMP system API. Response and status updates are propagated back to UWB through same integration layer

The overview diagram below illustrates the interaction between UWB, EMP Integration Service and EMP system services.



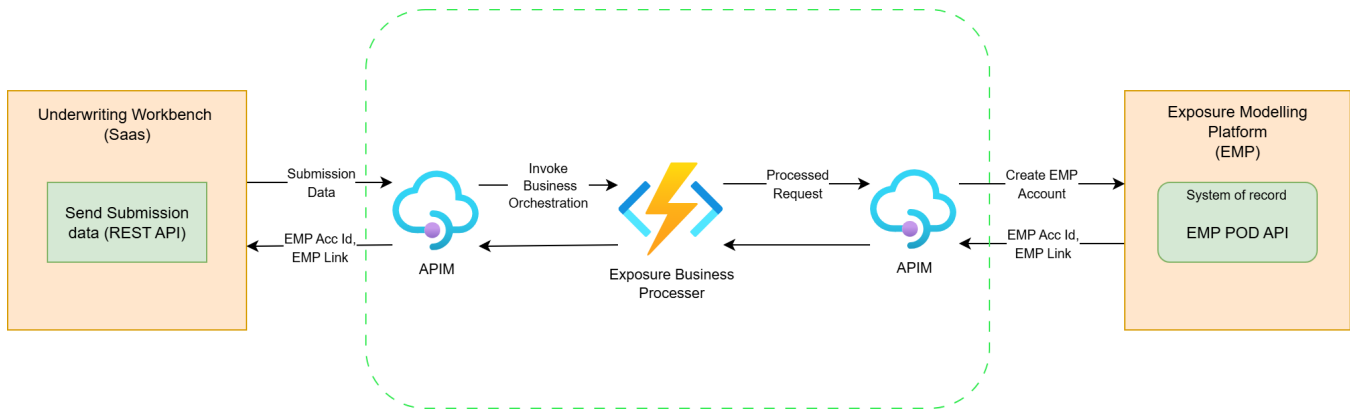
Key Interactions

#	Producer	Consumer	Interaction description and purpose
1	Underwriting Workbench (UWB)	APIM	WB triggers EMP request by sending submission data and metadata to APIM
2	APIM	Function App	APIM routes the validated request to the Function App for orchestration and processing
3	Function App	APIM	Function App retrieves routing, transformation and EMP specific config required to process the request. Function App sends transformed EMP compatible payload back to APIM for controlled outbound communication.
4	APIM	EMP Process API	APIM invokes EMP Process API to create and trigger the modelling request in EMP
5	EMP Process API	APIM	EMP returns process initiation status and EMP Link associated with the submission
6	APIM	Function App	APIM forwards EMP response back to Function App for post-processing and correlation.
7	Function App	Underwriting Workbench (UWB)	Function App Cascades the EMP process status and EMP link back to UWB via APIM

Design Diagram

Design considerations:

1. Azure API Management acts as secure integration gateway between Underwriting Workbench and EMP system services
2. The Integration between UWB and EMP uses a synchronous request-response pattern for request initiation.
3. WB requires acknowledgment the EMP request has been successfully triggered.
4. EMP returns: Process initiation status, EMP UI link tied to submission
5. EMP (Exposure Modelling Platform) is treated as System of Record for all modelling related data



1. Underwriting workbench triggers submission, which sends exposure submission data via a REST API call.
2. The request is securely transmitted over HTTPS (TCP 433) to Azure API Management (APIM).
3. APIM validates the incoming request (authentication, authorization and schema validation) and routes it to Exposure Business Processor.
4. Exposure Business Processor perform business orchestration and invokes the EMP POD API through APIM to create or update EMP account.
5. EMP POD API, hosted within the Exposure Modelling Platform, processes the request and persists data within EMP.
6. EMP returns a response containing EMP Account ID and EMP Link.
7. The response flows back through APIM to UWB, enabling users to navigate directly to the created EMP account.

The CDM Data payload from Underwriting Workbench

```

{
  "submission": {
    "submissionReference": "XL-S000010880",
    "targetSystem": "EMP",
    "submissionCurrencyCode": "EUR",
    "originatingOfficeCode": "OF0004",
    "lineOfOperationName": "JB-Manufacturing",
    "operationalSegmentName": "Downstream Energy",
    "systemAccountReference": "SX3000817503/161",
    "sourceCedentName": "Global Prog wCICLUK",
    "requestIdentifier": "54634-66543-8987",
    "overallStatusCode": "BND",
    "partyAssociations": [

```

```

{
  "partyRoleTypeCode": "Underwriter",
  "party": {
    "partyTypecode": "PERSON",
    "partyIdentifier": "X00026758",
    "displayName": "John Doe"
  }
},
{
  "partyRoleTypeCode": "Insured",
  "party": {
    "partyTypecode": "ORGANIZATION",
    "partyIdentifier": "X00012345",
    "displayName": "AXA XL"
  }
},
{
  "partyRoleTypeCode": "Broker",
  "party": {
    "partyTypecode": "ORGANIZATION",
    "partyIdentifier": "X00045678",
    "displayName": "ABC Corp"
  }
}
],
"programs": [
{
  "programNumber": "2424783",
  "policies": [
    {
      "policyReference": "XL-P000002567",
      "inceptionDate": "2024-01-01",
      "expirationDate": "2024-12-31",
      "directAssumedIndicator": "Direct"
    }
  ]
}
]
}
}

```

Mapping to EMP Payload

```

{
  "workbenchSubmissionId": "string", //Submission.SubmissionReference
  "accountName": "string", //PartyAssociation (partyRoleTypeCode = Insured)->PartyAssociation.Party.DisplayName //Mandatory
  "accountNumber": "string", //Submission.SystemAccountReference //Mandatory
  "inceptionDate": "yyyy-MM-dd", //Program[0].Policies[0].InceptionDate //Mandatory
  "expirationDate": "yyyy-MM-dd", //Program[0].Policies[0].ExpirationDate //Mandatory
  "obu": "string", //Submission.OpertionalSegmentName //Mandatory
  "legalEntity": "string", //Submission.SourceCededName
  "currencyCode": "string", //Submission.SubmissionCurrencyCode
  "lob": "string", //Submission.LineOfOperationName //Mandatory
  "underWriter": "string", //PartyAssociation (partyRoleTypeCode = Underwriter)->PartyAssociation.Party.DisplayName
  "cedant": "string", //Submission.TargetSystem //Conditional Mandatory
  "sovLink": "string", //Confussion around where to get this data from, need to confirm
  "broker": "string", //PartyAssociation (partyRoleTypeCode = Broker)->PartyAssociation.Party.DisplayName
  "branch": "string", //Submission.OriginatingOfficeCode
  "typeOfContract": "string", //Submission.Program[0].Policies[0].DirectAssumedIndicator
  "createdBy": "string", //need to hardcode //Mandatory
  "CATModelRequestId": "string", //Submission.RequestIdentifier
  "programStatus": "string" //Submission.OverallStatusCode //Mandatory
}

```

Azure Solution/Project

Solution, Function App and Repo information can be found here: [xlc-portfolio-tech-shared-services/AXAXL.Integration.UWWB.EMP](#)

Function Name: EmpProcessor

Function Trigger: HTTP trigger

API Name: ExposureModellingProcessor

Environment	APIM	Function Name
DEV	https://gateway-dev.api.axaxl.com/underwriting/empprocessor/v1/exposuremodelrequest	z-xlc-0339-uwwb-dv-ue2-fun13-uwwbintegrationslpl
SIT	https://gateway-test.api.axaxl.com/underwriting/empprocessor/v1/exposuremodelrequest	z-xlc-0339-uwwb-dv-ue2-fun14-uwwbintegrationslpl
UAT	https://gateway-qa.api.axaxl.com/underwriting/empprocessor/v1/exposuremodelrequest	z-xlc-0339-uwwb-pp-ue2-fun09-uwwbintegrationslpl
PPD	https://gateway-preprod.api.axaxl.com/underwriting/empprocessor/v1/exposuremodelrequest	z-xlc-0339-uwwb-pp-ue2-fun05-uwwbintegrationslpl
PROD	https://gateway-prod.api.axaxl.com/underwriting/empprocessor/v1/exposuremodelrequest	z-xlc-0339-uwwb-pr-ue2-fun02-uwwbintegrationslpl

Security

All integrations are secured using Azure AD-based authentication with token validation enforced at Azure API Management (APIM).

Communication between components occurs over HTTPS and access is restricted using RBAC and scoped permission to ensure secure and compliant data exchange.

Backend Services

API Name: ExposureModelingPlatform

Environment	APIM	EMP Backend
DEV	https://gateway-dev.api.axaxl.com/emp/exposuremodelingplatform/v1/api/Workbench	https://empcoreapi-dev.axaxl-cloud.com
SIT	https://gateway-test.api.axaxl.com/emp/exposuremodelingplatform/v1/api/Workbench	https://empcoreapi-dev2.axaxl-cloud.com/
UAT	https://gateway-preprod.api.axaxl.com/emp/exposuremodelingplatform/v2/api/Workbench	https://empcoreapi-qa2.axaxl-cloud.com/
PPD	https://gateway-preprod.api.axaxl.com/emp/exposuremodelingplatform/v1/api/Workbench	https://empcoreapi-qa.axaxl-cloud.com/
PROD	https://gateway-prod.api.axaxl.com/emp/exposuremodelingplatform/v1/api/Workbench	

Environment Mapping

UWB	APIM	Process API	EMP
PROD	PROD	PROD	PROD
PPD	PPD	PPD	PPD (QA1)
UAT	PPD	PPD	PPD (QA2)
DEV	DEV	DEV	NPD (DEV1)
SIT	TEST	TEST	NPD (DEV 2)
QA	TEST	TEST	NPD (DEV 2)
PROD(DR)	DR	DR	DR

Environment	UWB	Front End APIM (ExposureModellingProcessor)	Function	Process API -Backend API (ExposureModelingPlatform)	EMP
Dev	Dev	DEV (NPD) https://gateway-dev.api.axaxl.com/underwriting/empprocessor/v1/exposuremodelrequest	DEV z-xlc-0339-uwwb-dv-ue2-fun13-uwwbintegrationspl	DEV (NPD) https://gateway-dev.api.axaxl.com/emp/exposuremodelingplatform/v1/api/Workbench	NPD (DEV1) https://empcoreapi-dev.axaxl-cloud.com/
SIT	SIT	TEST (NPD) https://gateway-test.api.axaxl.com/underwriting/empprocessor/v1/exposuremodelrequest	SIT z-xlc-0339-uwwb-dv-ue2-fun14-uwwbintegrationspl	TEST (NPD) https://gateway-test.api.axaxl.com/emp/exposuremodelingplatform/v1/api/Workbench	NPD (DEV 2) https://empcoreapi-dev2.axaxl-cloud.com/
UAT	UAT	PPD -V2 https://gateway-preprod.api.axaxl.com/underwriting/empprocessor/v2/exposuremodelrequest	PPD z-xlc-0339-uwwb-pp-ue2-fun09-uwwbintegrationspl	PPD -V2 https://gateway-preprod.api.axaxl.com/emp/exposuremodelingplatform/v2/api/Workbench	PPD (QA2) https://empcoreapi-qa2.axaxl-cloud.com/
PPD	PPD	PPD-V1 https://gateway-preprod.api.axaxl.com/underwriting/empprocessor/v1/exposuremodelrequest	PPD z-xlc-0339-uwwb-pp-ue2-fun05-uwwbintegrationspl	PPD -V1 https://gateway-preprod.api.axaxl.com/emp/exposuremodelingplatform/v1/api/Workbench	PPD (QA1) https://empcoreapi-qa.axaxl-cloud.com/
PRD	PRD	PRD	PRD z-xlc-0339-uwwb-pr-ue2-fun02-uwwbintegrationspl	PRD	PRD
DR	DR		DR		
QA		TEST (NPD)	TEST z-xlc-0339-uwwb-dv-ue2-fun14-uwwbintegrationspl	TEST (NPD)	NPD (DEV 2) https://empcoreapi-dev2.axaxl-cloud.com/

Reference

EMP Cloud LLD: [Underwriting Workbench-Salesforce and EMP - Cloud Infrastructure Low Level Design - RADAR - Confluence by AXA GO](#)

[Underwriting Workbench-Salesforce and EMP - Scope Page - RADAR - Confluence by AXA GO](#)

Business Catalogue: [CDM Business Catalogue.xlsb](#)

Logical Data Model: [LDM_Enterprise_UnderwritingWorkbench_EMP_CDM_V1.pdf](#)

Canonical Data Model (Json): [Canonical Data Model V10](#) (relevant JSON for EMP– submission, party, policy)

CDM Version History: <https://confluence.axa.com/confluence/display/XLCDIADEEP/CDM+Version+History>

Please refer Col G and H for the EMP and CDM fields mapping [Business Catalogue_UWB_EMP V1.0.xlsx](#).